Systems Engineering - Midterm Review

Team 9 - Semantic Transmitter - "Semantic Audio Communication using GenAI"

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1. Introduction
   - Team Organization
   - Challenge Description
2. Validated Requirements
3. Outcome of the Market Survey
4. System Concept
5. System Breakdown Structure
6. Functional Architecture
7. Work Breakdown Structure + Gantt
8. Measures of Effectiveness and Performance
9. Risk Management
10. Conclusion
    - Internal Evaluation
    - Next Steps

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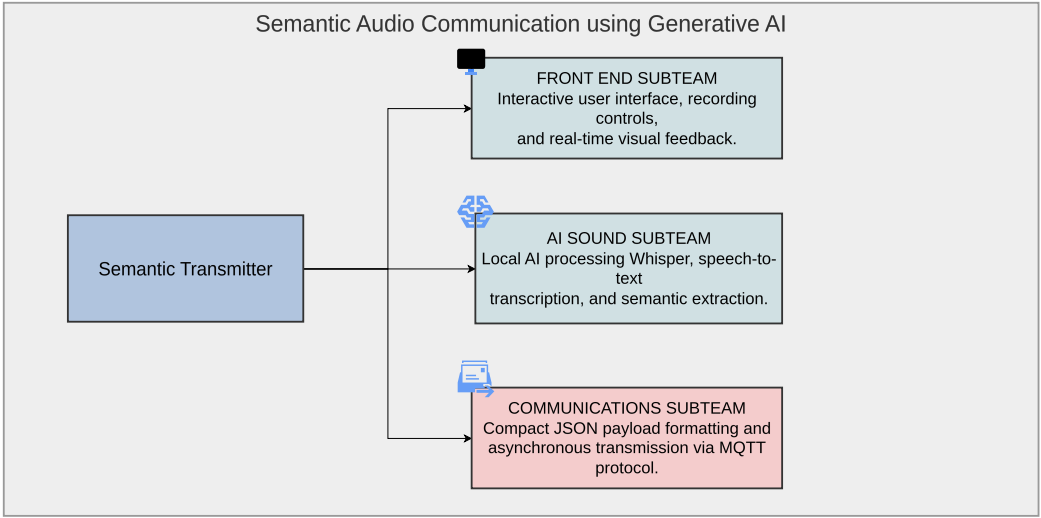
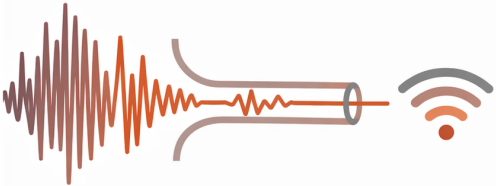
Team organization



Challenge Description

The Problem (Status Quo)



Header: Traditional Voice Messaging

* High bandwidth consumption.
* Severe delivery delays on unstable links.

The Challenge/Our Solution

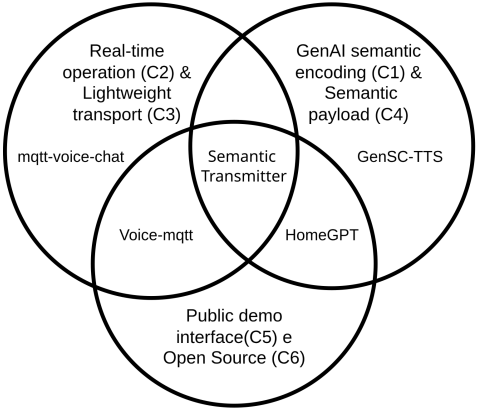
Header: Semantic Transmission

* Edge AI local meaning extraction.
* Smaller payload containing semantic information.
* Capable of working in unstable network conditions.

Validated Requirements

Category	Validated requirements
Communication	Asynchronous message sending
	Standardized semantic payload with transcript and metadata
	Delivery-state feedback
	Measurable bandwidth reduction vs. raw audio transmission
User Interface	Explicit start/stop recording controls
	Dynamic transcript display after processing
	Real-time status feedback
	Graceful, non-disruptive error handling during demos
AI Model	Model runs locally; retry path for capture/transcription failures; transcript shown before transmission; raw audio discarded after transcription; low end-to-end latency
Validation concept	Peer Reviews & Checklist Comparison

Outcome of the Market Survey



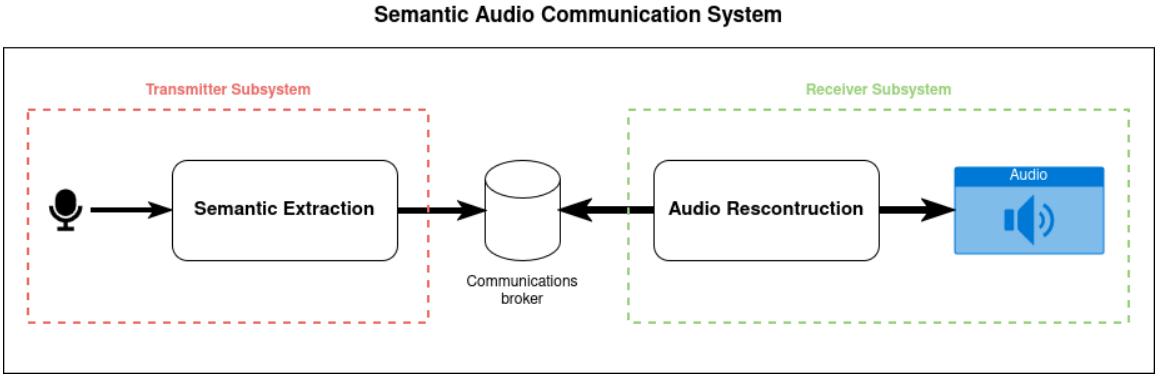
AI PROCESSING: Whisper large-v3

- High Accuracy STT & Native Portuguese Support.
- 100% Offline Capable (Critical for live demos)
- Free & Open-Source

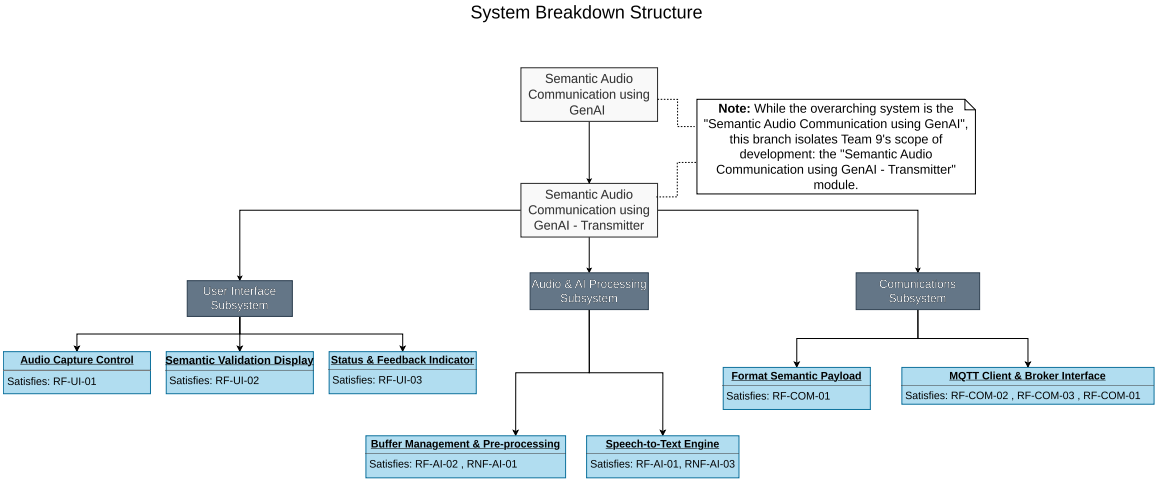
COMMUNICATION TRANSPORT: MQTT

- Minimal network overhead & very low latency
- Pub/Sub Broker Paradigm
- Asynchronous delivery (Decouples TX from RX)

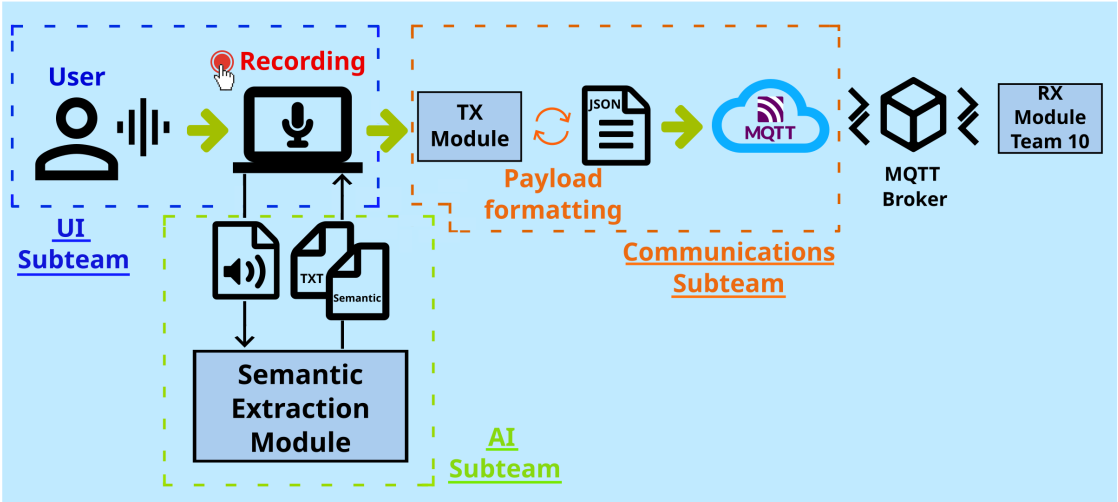
System Concept



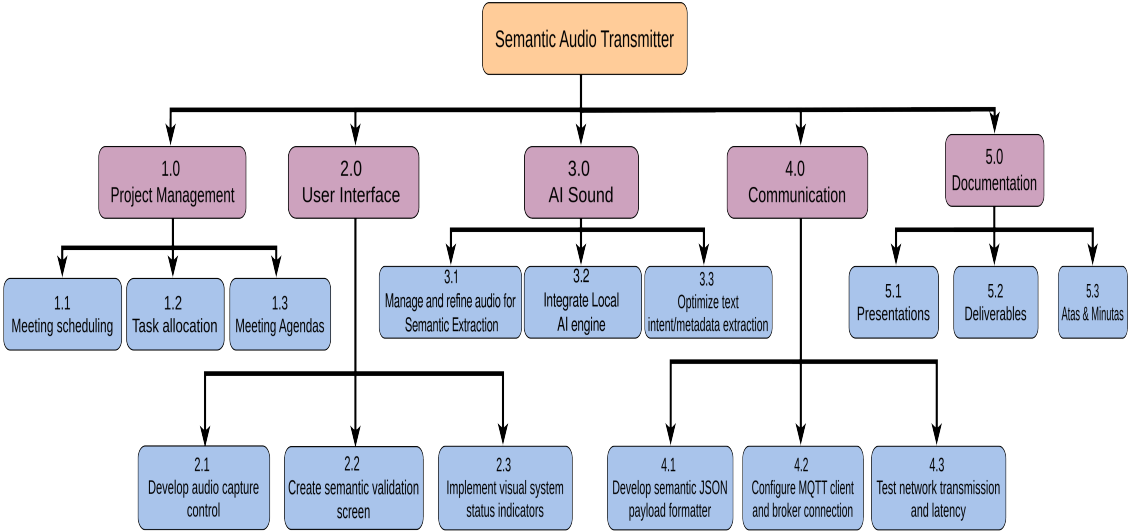
System Breakdown Structure



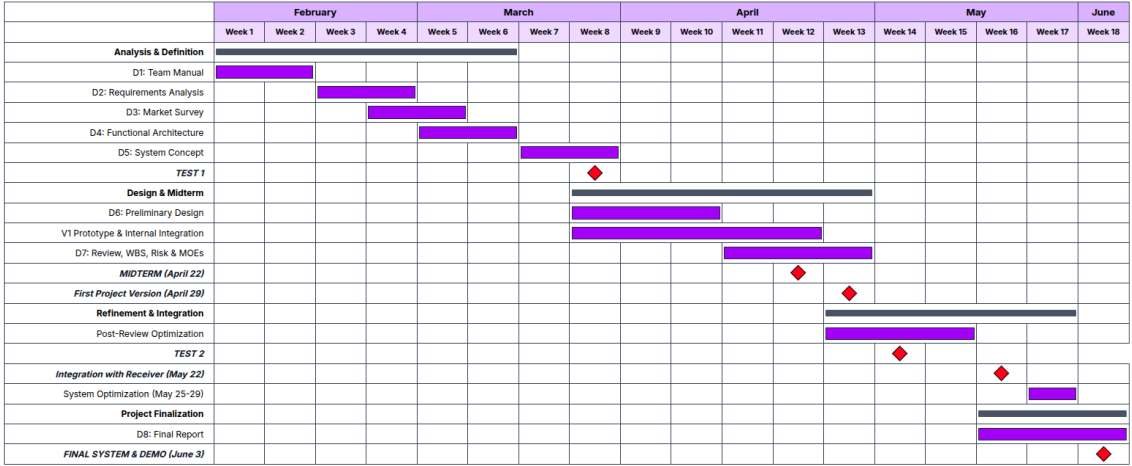
Functional Architecture



Work Breakdown Structure + Gantt



Gantt Chart


February
March
April
May
June
Week 1
Week 2
Week 3
Week 4
Week 5
Week 6
Week 7
Week 8
Week 9
Week 10
Week 11
Week 12
Week 13
Week 14
Week 15
Week 16
Week 17
Week 18
Analysis & Definition
D1: Team Manual
D2: Requirements Analysis
D3: Market Survey
D4: Functional Architecture
D5: System Concept
TEST 1
Design & Midterm
D6: Preliminary Design
V1 Prototype & Internal Integration
D7: Review, WBS, Risk & MOEs
MIDTERM (April 22)
First Project Version (April 29)
Refinement & Integration
Post-Review Optimization
TEST 2
Integration with Receiver (May 22)
System Optimization (May 25-29)
Project Finalization
D8: Final Report
FINAL SYSTEM & DEMO (June 3)

Measures of Effectiveness and Performance

WER

Semantic Accuracy

- Evaluates the system's effectiveness in capturing the users's correct intent.
- If the WER is high, the main objective semantic communication) fails.

Delivery Reliability

- The percentage of payload that are successfully published to the Broker out of the total number of attempts.
- The percentage of payloads that are compliant with the schema chosen to format the payload (JSON).

End-to-End Latency

[MQTT Delivery Reliability]

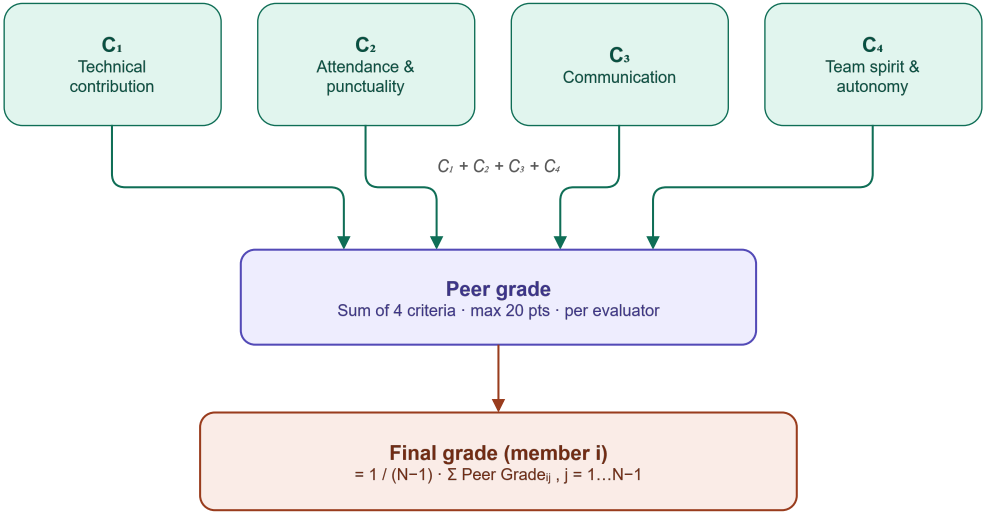
- The total time measured from the moment the user finishes speaking until the formatted payload is sent.
- The risk report stipulates that the model's latency cannot exceed the defined limit.

Risk Management

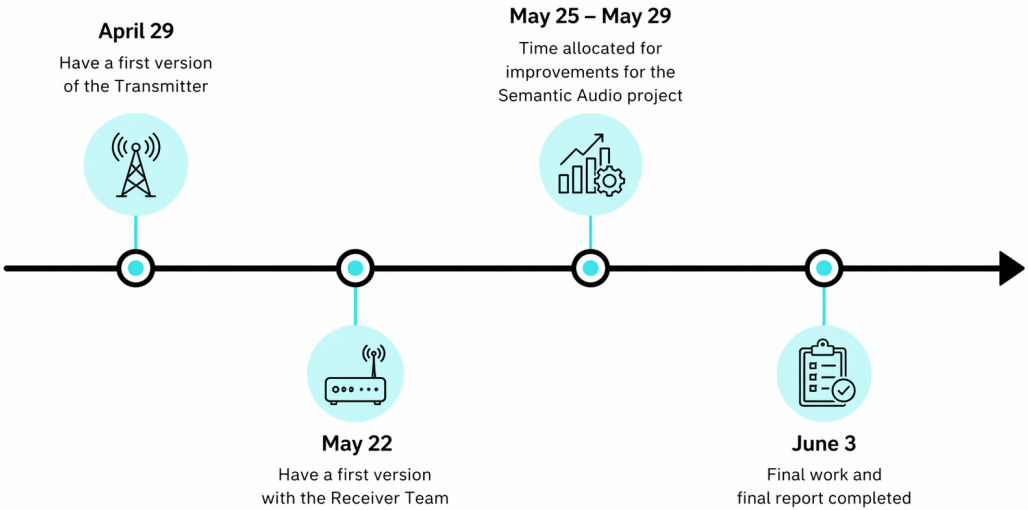
PoO \ Impact	Low	Medium	High
High			High Latency From AI Model
Medium		Insufficient Audio Quality	MQTT Communications Failures
Low		Resource Exhaustion During Demos	HW Failures

PoO - Probability of Occurence

Internal Evaluation



End of Semester Plan



Thank You for Your Attention!

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[7] Team 9.
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[8] Team 9.
Deliverable 6: Preliminary design — semantic audio communication using genai – transmitter.
Technical Report Deliverable 6, Faculty of Engineering, University of Porto, April 2026.

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[9] Team 9.
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